

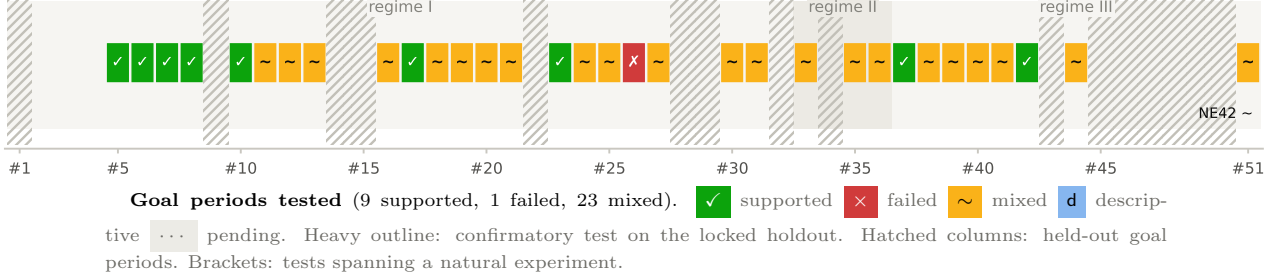
H34: Idea cascades follow a power law whose exponent reads off the distance to criticality

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-05 (round 2)

Complete 65% Faithfulness 2.5/5 Usefulness 2.5/5

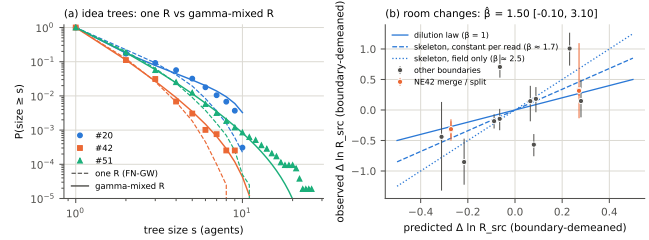
Bottom line. Idea branching is heterogeneous and shared along trees: a gamma mixture of idea-level ratios ($CV^2 \approx 1$) fits the heavy tail that one ratio misses, adopters out-spread inventors 1.5:1, and in a room merge and split branching reversed with room size.

level: descriptive · pipeline checks 86% · updated 2026-10-05 (round 2)



WHAT HAPPENED

H34 builds exposure trees of new ideas in 32 non-reserved periods; an agent's first use gets as parent the latest earlier use it read. Rounds 1–1b: cascades are subcritical everywhere ($\hat{R} = 0.09\text{--}0.40$), tails are heavier than one branching law, and half the exposure lock is convergence. Round 2: (R1) a gamma mixture of idea-level branching ratios fits the tail in 28/32 periods (one ratio: 9/32), and adopters out-spread inventors. (R4) Paraphrase ideas (two embedding models) are subcritical at half the marker value, but their adoption is mostly convergence. (R2) Across the NE42 merge and split, branching reverses with room size; per-read dilution is not identified. No reserved-data run.



Observables. (a) Fraction of exposure trees that reach at least s agents in #20, #42 and #51. Dashed: one branching ratio (FN-GW). Solid: the gamma mixture of idea-level ratios. The mixture follows the tail; one ratio falls short beyond $s \approx 4$. (b) Room changes at six goal boundaries, after removing each boundary's common shock. Orange: the NE42 merge and split. Lines: the dilution law ($\beta = 1$), and synthetic worlds on the real timelines with a constant chance per read ($\beta \approx 1.7$) or a field only ($\beta \approx 2.5$). The data cannot tell them apart.

PHYSICAL MODEL

Contagion (model 03) runs as a Galton–Watson cascade (model 09) in a room of N agents with depletion (FN-GW). The degree of freedom is an agent's first use of an idea. Each idea i has its own ratio $R_{0,i}$, gamma-distributed with mean μ , shape α ($CV^2 = 1/\alpha$). $P_{\text{FN}}(s | R_0)$ is the exact size law. ρ_{off} : non-root over root offspring. R_{src} : a group's offspring per first use; β scales the room-size rule.

$$\hat{R} = \frac{\# \text{ first uses with an agent parent}}{\# \text{ agent first uses}}, \quad R_{0,i} \sim \Gamma(\alpha, \mu/\alpha)$$

$$P(\{s_{ij}\}_j) = \int dR_0 \Gamma(R_0; \alpha, \mu/\alpha) \prod_j P_{\text{FN}}(s_{ij} | R_0)$$

$$\Delta \ln R_{\text{src}} = \beta \cdot 0.451 \ln \frac{N_a - 1}{N_b - 1} + \text{boundary effect}$$

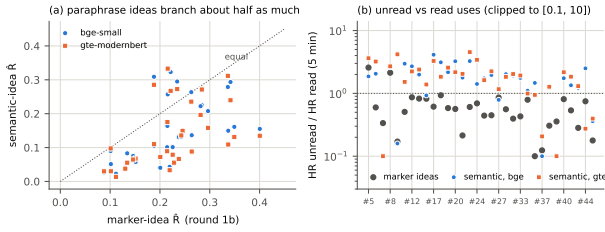
SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

For tail risk, use the two-parameter mixture (mean ≈ 0.2 , $CV^2 \approx 1$); one ratio under-predicts ideas that reach five agents. An idea's second-generation holders spread it 1.5 times as often as its inventor: watch them. Measure copying by reuse of names, links and numbers; paraphrase similarity mostly measures a shared stimulus. Merging rooms widens each agent's reach and splitting reverses it.

H34-idea-cascades · generated 2026-10-07 17:50 by in-
fra/summaries/build_summaries.py · card hypotheses/H34-idea-
cascades/README.md · ratings: H34 agent (suggested); round 1b RE-D1
(suggested); round 2 agent (suggested, 2026-10-05)

RESULTS VS PREDICTIONS

Prediction	Observed	Verdict
R1-P1 mixture covers the tail	28/32 (one ratio: 9/32)	pass
R1-P2 heterogeneity (LR)	31/32; median $\hat{\alpha} = 0.96$ (0.27–5.3)	pass
R1-P3 beats beta-binomial	21/32 day-ahead, gain mostly from #51	pass (small)
R1-P4 unfitted ρ_{off}	1.52, > 1 in 31/32; homogeneous references ≤ 0.91	pass
R4-P1 paraphrase ideas subcritical	$\hat{R}$ 0.13 (bge), 0.11 (gte), CI < 1 in 32/32; 0.39 at a looser threshold	pass
R4-P2/P3 rank and model agreement	ρ with markers 0.56–0.63; bge vs gte 0.94	pass
R4-P4 read beats unread	7/32	fail
R4-P5 more convergence than markers	unread/read 1.8–1.9 vs 0.56	pass
R2-P2 NE42 reversal $D > 0$	+1.29 [0.74, 2.11]; predicted +1.09	pass
R2-P1 dilution slope β	1.50 [−0.10, 3.10]	open



Second figure. (a) One dot per period and model. Paraphrase ideas (leader clustering at rate-matched thresholds) branch about half as much as marker ideas and rank the periods alike. (b) Adoption hazard after another agent's use that was posted but not yet read, over the hazard after one already read, both within 5 min. For markers (gray) read uses win. For paraphrases (blue, orange) unread uses win, so their adoption is mostly convergence.

GOAL PERIODS IN DETAIL

G05 ✓ 1b: supported (ledger visibility: R 0.22, HR₁₀ 5.8 [...]
G06 ✓ 1b: supported (ledger visibility: R 0.11, HR₁₀ 5.6 [...]
G07 ✓ 1b: supported (ledger visibility: R 0.10, HR₁₀ 16.1 ...
G08 ✓ 1b: supported (ledger visibility: R 0.13, HR₁₀ 1.7 [...]
G10 ✓ 1b: supported (ledger visibility: R 0.19, HR₁₀ 9.7 [...]
G11 ~ 1b: mixed (ledger visibility: R 0.21, HR₁₀ 12.2 [6.7...
G12 ~ 1b: mixed (ledger visibility: R 0.28, HR₁₀ 6.2 [4.7,...
G13 ~ 1b: mixed (ledger visibility: R 0.15, HR₁₀ 4.4 [3.0,...
G16 ~ 1b: mixed (ledger visibility: R 0.10, HR₁₀ 3.9 [2.3,...
G17 ✓ 1b: supported (ledger visibility: R 0.25, HR₁₀ 11.5 ...
+23 more periods: see the card.

FAITHFULNESS SCORECARD

A mapping	1	F identifiability	2
B assumptions	1	G ground truth	1
C adequacy	2	H comparative	2
D unfitted predictions	2	I transfer	1
E interventional	1		

Level: **descriptive**. Scale 0 not done or failed, 1 partial, 2 passed (`writeup/paper.tex`).

CAVEATS

Tail fit and likelihood ratio cannot tell idea-level from node-level spread; only the non-root/root ratio does. Most ideas have one tree, so idea and seeding-moment heterogeneity are not separated. The forecast gain is mostly #51. Paraphrase $\hat{R}$ depends on the threshold (0.008–0.39). Unread uses are more recent, which overstates convergence. Each room change is also a goal change, and worlds without per-read dilution also predict the reversal.

ROUND 2

Per-read dilution inside #51 with a power check; ideas with several trees to separate idea from seeding moment; re-freeze the confirm script with the mixture clauses.

- **Essence.** how far does a new idea or claim spread in an agent swarm, and what sets it: exposure, salience or a shared field?
- **Round-3 redirects (newest).**
- **H34-R5** Per-read dilution with a cleaner dose. Use #51's day-level room sizes and roster joins (NE33) with a within-room, within-goal design, so that goal shocks do not swamp β .
- **H34-R6** Idea vs tree heterogeneity. Ideas with several independent trees (several roots) test whether R is a property of the idea or of the seeding moment: the mixture predicts correlated tree sizes within an idea.
- **H34-R7** Re-freeze confirm.py on ledger visibility with the Γ -FN tail clause ($C6 \rightarrow \text{"}\Gamma\text{-FN covers } P(s \geq 5)\text{"}$) and the non-root/root ratio ≤ 1 clause.
- **H34-R3** (unchanged). Claims proper, with Jev labels (needs paid labels).

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